Unequal-Area Facility Layout: Computational Study (benchmark v1)

October 12, 2026

1 Method

The two-stage method (slicing-tree annealing, then sequence-pair search with exact SOCP geometry) is described in the method section of the main paper.

2 Computational experiments

This section compares the two-stage method with five heuristic baselines and an exact convex MINLP on five unequal-area facility layout (UA-FLP) instances from the literature. All methods solve the same model: rectangular departments with exact areas, a maximum aspect ratio α, a fixed rectangular floor, no overlap, and the objective $\sum_{i<j} w_{ij}\,(|x_i - x_j| + |y_i - y_j|)$ on department centroids.

2.1 Test instances

Table 1 lists the instances. O7, vC10a, AB20 and Du62 fill the floor completely (Du62 up to rounding of the floor side), so every feasible layout is a partition of the floor. SC30 leaves 9.44 % of the floor free; empty space is allowed in our model. The floors and aspect limits are those used by Jeong and Chae [1]; for Du62 we use the 117.124×117.124 square on which they report their results. O7 is also instance `o7` of MINLPLib [2], which takes it from Westerlund and Lundqvist [3]; MINLPLib lists the global optimum 131.6531381, with dual bounds of SCIP and BARON equal to this value. Two data details matter for the comparison with published values. First, vC10a uses $w_{9,10} = 59.2$; this is the value under which the layout reported in [1] has cost 18,522.79. Second, the AB20 flow matrix is the one of the Kang and Chae data file, which equals the matrix of Armour and Buffa [4] multiplied by 10 (with the area and floor scaled accordingly).

2.2 Methods and settings

All stochastic baselines, and Stage 1 of our method, work on the slicing-tree (STS) encoding of Kang and Chae [9] and Jeong and Chae [1]: a department sequence, a cut order and a cut code per cut. Cuts

Table 1: Test instances and time budgets. n: number of departments; α: maximum aspect ratio; fill: total department area divided by floor area; pairs: department pairs with non-zero flow; budget: wall-clock limit per run for every stochastic method.

Instance	Origin	n	Floor $W \times H$	α	Fill (%)	Pairs	Budget (s)
O7	[5]	7	8.54×13	4	99.98	10	20
vC10a	[6]	10	25×51	5	100.00	12	20
AB20	[4]	20	30×20	4	100.00	62	60
SC30	[7]	30	15×12	5	90.56	50	60
Du62	[8]	62	117.124×117.124	4	100.00	1182	150

split the current rectangle in proportion to the areas of the two parts, so areas are exact. For SC30 the free area is split into 17 dummy blocks of equal area, without flow and without a shape limit, as in the Kang and Chae data file. The STS methods use the same static penalty $\pi = 0.05\,(\sum_{i<j} w_{ij})(W+H)$ per unit of aspect excess, and they return the best layout that meets the aspect limit exactly. Slicing layouts cannot represent every layout; on O7 the optimal layout is not a slicing layout.

Construct. A deterministic construction heuristic. Departments are ordered greedily by flow to the departments already ordered (a CORELAP-style rule on the flow matrix), then placed in a flexible bay structure. All bay counts $k = 1, \ldots, n$ and both bay orientations are tried; the best layout that meets the aspect limit is returned. One run per instance.

Swap. CRAFT-style pairwise exchange [4] on the STS sequence: steepest descent over all pairwise swaps, restarted from a random genome at each local optimum.

GA. A genetic algorithm without local search: population 50, two elites, binary tournament, PMX on the sequence and the cut order, one-point crossover on the cut codes, mutation probability 0.2.

SA. Simulated annealing with three move types; the start temperature accepts a median worsening move with probability 0.75, and the temperature falls geometrically in elapsed time to 10^{-4} of its start value.

CRO. Our reproduction of the coral reefs optimisation of Jeong and Chae [1] (grid 7×7, parameters of their Table 5, duplicate larvae rejected). The paper's stopping rule is replaced by the common time budget. Run to the paper's stopping rule, our reproduction reaches the published best value on O7 in 30/30 runs and on vC10a in 22/30 runs; on AB20 (7×7) its mean is 5,425.39 against the published 5,472.09, on Du62 its mean is 3,698,739.27 against 3,741,596.01, and on SC30 it is not reproduced (mean 4,331.87 against 3,877.63). "CRO" below always means our reproduction.

SA + polish, CRO + polish. A fairness control: the final layout of each SA and CRO run is converted into a sequence pair and solved once by the Stage-2 SOCP of our method (the same model and margins). The SOCP cannot make a layout worse; if it gives no gain beyond 10^{-8} relative, the original layout is kept. The polish took at most 0.17 s (SA) and 0.15 s (CRO) per run and is not counted in the budget. These rows separate the gain from the SOCP geometry from the gain from a better search.

MINLP. An exact convex mixed-integer nonlinear model solved by SCIP 10 (one thread, 300 s): continuous positions and sizes, the area constraint $h_i \geq A_i/w_i$ (convex), linear aspect constraints, four binaries per department pair with big-M disjunctions, and absolute-value distances. Its optimum is the UA-FLP optimum (up to the solver feasibility tolerance 10^{-7}), and its dual bound is a valid lower bound.

Ours. The two-stage method of Section 1. Stage 1 is slicing-tree annealing for half of the budget and keeps an elite pool of up to four layouts. Stage 2 converts each elite layout into a sequence pair [10, 7] and searches over sequence pairs; each sequence pair is evaluated by a convex second-order cone program (SOCP) solved by Clarabel [11]. If fewer than 3000 SOCP solves fit in the remaining time, Stage 1 continues to the end of the budget and Stage 2 is a single SOCP solve (polish).

We also ran two restricted exact models, one run of 300 s each: a MIP in which every department chooses one of at most nine fixed exact-area shapes (SCIP through OR-Tools [12]), and a CP-SAT model [13] on an integer grid. Both are infeasible by construction on the full floors (no tiling with so few shapes; grid rounding leaves fewer cells than the areas need on O7 and Du62), and neither found a feasible layout on any instance. We do not report them further.

The parameters of Swap, GA, SA and CRO are our own choices or the published ones and were not tuned on these instances. The Stage-1 cycle length and temperatures of our method were tuned on AB20 and SC30, and the Stage-2 temperatures on SC30; both are test instances (Section 2.7).

2.3 Protocol

Budgets and seeds. Every stochastic method gets the same wall-clock budget per run (Table 1): 20 s for O7 and vC10a, 60 s for AB20 and SC30, and 150 s for Du62. Each stochastic method is run with seeds 0–9. The construction heuristic and the exact models are run once.

Machines. All stochastic methods, including all runs of our method and its variants, ran on the same machine: a Linux (aarch64) virtual machine with 4 cores, Python 3.10.12, NumPy 2.2.6 and Numba 0.68.0, with three runs in parallel, each as a separate process. The exact models (MINLP, MIP, CP-SAT) ran on a different machine with 2 cores (Intel Xeon, 2.8 GHz), one run at a time. The polish of the SA and CRO layouts was applied afterwards to the stored layouts, and its time is recorded separately.

Budget overruns. The runs of our method and its variants were made before a fix of the time check in Stage 1 and can exceed the budget. The largest overrun is 0.14 s (0.7 %) on O7, 0.17 s (0.9 %) on vC10a, 0.24 s (0.4 %) on AB20, 0.71 s (1.2 %) on SC30 and 4.28 s (2.9 %) on Du62. The baselines exceed the budget by at most 0.02 s.

Independent checker and tolerance. No method reports its own cost. After each run, the layout is checked by an independent evaluator, which recomputes the cost from the rectangles and checks every constraint with tolerance $\varepsilon = 10^{-6}$: $w_i h_i \geq A_i(1-\varepsilon)$, aspect ratio at most $\alpha+\varepsilon$, containment in the floor up to ε, and no pairwise overlap larger than ε in both directions. A run counts as feasible only if the evaluator accepts its layout. A second, separate verification script re-checked all stored layouts: layouts produced by a solver (SOCP, MINLP) meet the limits to about 10^{-7}, and slicing layouts are exact to about 10^{-9}. All 538 stored layouts of the protocol were checked again at the end; stored and recomputed cost and feasibility agree for every file. The code of each round of runs is identified by MD5 checksums, and the test suite of the evaluator and models passed (42 tests passed, 3 expected failures).

Smoke tests. Before the main runs, every heuristic was run for a few seconds on O7, vC10a and AB20, and all produced feasible layouts. The construction heuristic first had no feasible layout on AB20 with vertical bays only, which is why horizontal bays were added. The MINLP model was validated on O7 with a 3600 s run: SCIP proved optimality at 2322 s with value 131.65313440, which matches the MINLPLib optimum up to the solver tolerance.

Reference values and statistics. Gaps are $100\,(\text{cost}-\text{ref})/\text{ref}$ with the reference values in Table 2: O7 131.6531381 (proven optimum, MINLPLib [2]); vC10a 18,522.79 and SC30 3,343.47 from Jeong and Chae [1, Table 6]; AB20 5,021.17 from Gonçalves and Resende [14, Table 12]; Du62 3,635,307.30 from Kang and Chae [9] as listed in [1, Table 6]. Section 2.5 discusses how far each value is comparable. To compare methods we use two-sided Mann–Whitney U tests on the ten costs per method (the seeds are independent random streams, so the samples are unpaired), with Holm's correction over the comparisons within an instance.

2.4 Results

Table 2 gives the results, Figure 1 shows the gap of every run, and Table 3 gives the tests.

Feasibility. All heuristics found feasible layouts in every run, with one exception: Swap found a feasible layout on Du62 in only 1 of 10 runs. The MINLP found the optimum of O7 (131.6531381, first reached after 172 s) but could not prove it within 300 s (gap 36.2 %). On vC10a and SC30 its best layouts are far from the others (22,430.69 and 10,073.72), and on AB20 and Du62 it found no feasible layout. Its lower bounds after 300 s are weak (for example 711.41 on AB20).

Solution quality. Our method has the lowest mean cost on all five instances; on vC10a, SA, CRO and their polished versions have the same mean (every run of these methods ends at the published layout, 18,522.79). On O7 every run of our method reaches the optimum up to our area safety margin, while Swap, SA and CRO end at the best slicing layout 131.69 in every run; one polish of their layouts closes most of this gap (mean gap 0.003 % for SA + polish against 0.000 % for our method, Table 2). Against the strongest baseline, SA, the difference is significant on SC30 (adjusted $p = 2.3 \cdot 10^{-3}$) but not on AB20

Table 2: Results under the common protocol (10 seeds per stochastic method; one run for the deterministic construction heuristic and one 300 s run for the exact convex MINLP). SA/CRO + polish: one Stage-2 SOCP solve applied to the final layout of each SA/CRO run (polish time, at most 0.17 s and 0.15 s, not included in the budget). Feas.: runs with a feasible layout / runs. Costs are those of the independent evaluator, over feasible runs only; std is the sample standard deviation. Gaps are $100\,(\text{cost} - \text{ref})/\text{ref}$ for the best run and for the mean; the O7 reference is the proven optimum 131.6531381. Bold: lowest mean among the methods with 10 runs.

Method	Feas.	Best	Mean ± std	Gap best (%)	Gap mean (%)
O7 (reference 131.6531, budget 20 s)					
Construct	1/1	151.0684	151.0684	14.747	14.747
Swap	10/10	131.6892	131.6892 ± 0.0000	0.027	0.027
GA	10/10	131.6892	137.7969 ± 3.7760	0.027	4.667
SA	10/10	131.6892	131.6892 ± 0.0000	0.027	0.027
SA + polish	10/10	131.6532	131.6565 ± 0.0030	0.000	0.003
CRO	10/10	131.6892	131.6892 ± 0.0000	0.027	0.027
CRO + polish	10/10	131.6532	131.6570 ± 0.0020	0.000	0.003
MINLP	1/1	131.6531	131.6531	0.000	0.000
Ours	10/10	131.6532	**131.6532** ± 0.0000	0.000	0.000
vC10a (reference 18,522.79, budget 20 s)					
Construct	1/1	28,322.56	28,322.56	52.91	52.91
Swap	10/10	18,522.79	18,804.48 ± 356.46	0.00	1.52
GA	10/10	19,670.20	22,007.37 ± 1,956.26	6.19	18.81
SA	10/10	18,522.79	**18,522.79** ± 0.00	0.00	0.00
SA + polish	10/10	18,522.79	**18,522.79** ± 0.00	0.00	0.00
CRO	10/10	18,522.79	**18,522.79** ± 0.00	0.00	0.00
CRO + polish	10/10	18,522.79	**18,522.79** ± 0.00	0.00	0.00
MINLP	1/1	22,430.69	22,430.69	21.10	21.10
Ours	10/10	18,522.79	**18,522.79** ± 0.00	0.00	0.00
AB20 (reference 5,021.17, budget 60 s)					
Construct	1/1	8,824.29	8,824.29	75.74	75.74
Swap	10/10	5,495.96	5,573.29 ± 56.65	9.46	11.00
GA	10/10	5,537.01	6,216.97 ± 519.12	10.27	23.82
SA	10/10	4,926.92	5,043.20 ± 68.24	−1.88	0.44
SA + polish	10/10	4,926.92	5,043.20 ± 68.24	−1.88	0.44
CRO	10/10	5,123.24	5,291.01 ± 125.85	2.03	5.37
CRO + polish	10/10	5,123.24	5,291.01 ± 125.85	2.03	5.37
MINLP	0/1	–	–	–	–
Ours	10/10	4,949.54	**5,020.81** ± 47.79	−1.43	−0.01
SC30 (reference 3,343.47, budget 60 s)					
Construct	1/1	9,880.40	9,880.40	195.51	195.51
Swap	10/10	4,386.11	4,534.38 ± 136.84	31.18	35.62
GA	10/10	4,632.73	5,124.95 ± 291.64	38.56	53.28
SA	10/10	3,469.66	3,646.64 ± 142.30	3.77	9.07
SA + polish	10/10	3,381.58	3,534.20 ± 108.68	1.14	5.70
CRO	10/10	3,734.94	4,311.81 ± 301.61	11.71	28.96
CRO + polish	10/10	3,528.51	4,126.31 ± 300.51	5.53	23.41
MINLP	1/1	10,073.72	10,073.72	201.30	201.30
Ours	10/10	3,301.27	**3,407.85** ± 69.18	−1.26	1.93
Du62 (reference 3,635,307.30, budget 150 s)					
Construct	1/1	4,684,133.69	4,684,133.69	28.85	28.85
Swap	1/10	4,023,859.84	4,023,859.84	10.69	10.69
GA	10/10	3,696,431.38	3,735,359.99 ± 23,459.65	1.68	2.75
SA	10/10	3,623,772.52	3,666,052.33 ± 26,524.87	−0.32	0.85
SA + polish	10/10	3,623,751.81	3,666,030.22 ± 26,533.57	−0.32	0.85
CRO	10/10	3,678,297.88	3,700,566.61 ± 17,186.34	1.18	1.80
CRO + polish	10/10	3,678,291.91	3,700,540.99 ± 17,198.82	1.18	1.79
MINLP	0/1	–	–	–	–
Ours	10/10	3,629,508.93	**3,637,888.92** ± 5,454.26	−0.16	0.07

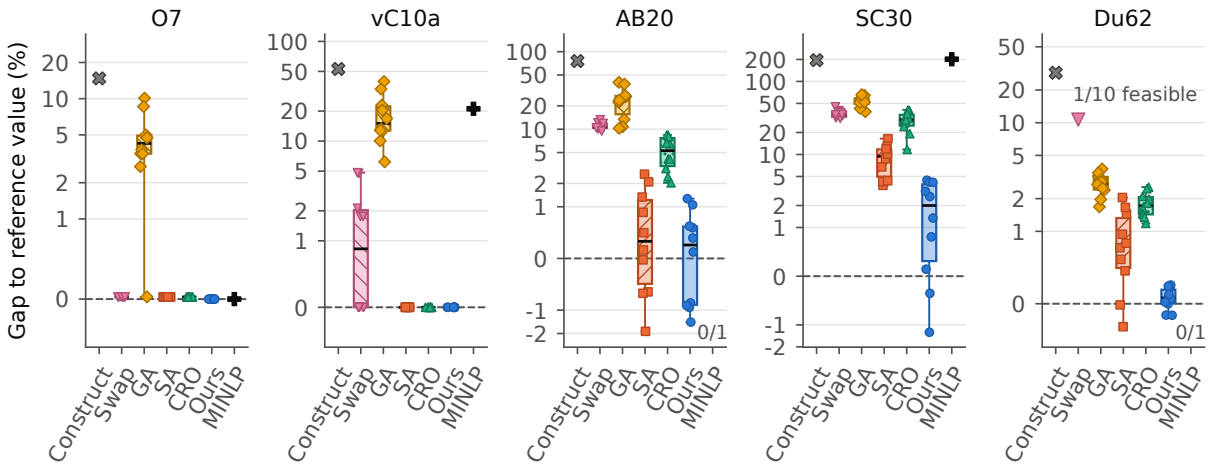


Figure 1: Gap of each run to the reference value (symmetric log scale; linear between -1 and 1 %). Boxes show quartiles and whiskers the range over the feasible runs; every run is also drawn as a point. Construct and MINLP are single runs; "0/1": no feasible layout. Annotation above a box: number of feasible runs when it is below 10. References: O7, the proven optimum (MINLPLib [2]); vC10a, SC30 from Jeong and Chae [1]; AB20 from Gonçalves and Resende [14]; Du62 from Kang and Chae [9] as cited in [1]. The polish rows are in Table 2.

Table 3: Two-sided Mann–Whitney U tests of the full method against each other method (10 runs each, unpaired), Holm-adjusted within each instance over the eight comparisons. Each cell: adjusted p-value and, in brackets, the effect size $A_{12} = P(\text{ours} < \text{other}) + \frac{1}{2}P(\text{tie})$ (values above 0.5 favour our method). Bold: adjusted $p < 0.05$. tie: all 20 costs equal within 10^{-6} relative. n/a: fewer than two feasible runs. The p-values use the exact distribution when there are no ties and the normal approximation otherwise.

Ours vs	O7	vC10a	AB20	SC30	Du62
Swap	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	$\mathbf{1.3 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	n/a
GA	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	$\mathbf{1.3 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.3 \cdot 10^{-3}}$ (1.00)
SA	$\mathbf{7.6 \cdot 10^{-4}}$ (1.00)	tie	1 (0.61)	$\mathbf{2.3 \cdot 10^{-3}}$ (0.95)	0.069 (0.82)
SA + polish	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	tie	1 (0.61)	**0.023** (0.84)	0.069 (0.82)
CRO	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	tie	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.3 \cdot 10^{-3}}$ (1.00)
CRO + polish	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	tie	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.5 \cdot 10^{-3}}$ (1.00)	$\mathbf{1.3 \cdot 10^{-3}}$ (1.00)
Stage 1 only	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	tie	1 (0.39)	$\mathbf{1.8 \cdot 10^{-3}}$ (0.97)	1 (0.42)
Stage 1 + polish	$\mathbf{7.8 \cdot 10^{-4}}$ (1.00)	tie	1 (0.41)	**0.023** (0.83)	1 (0.41)

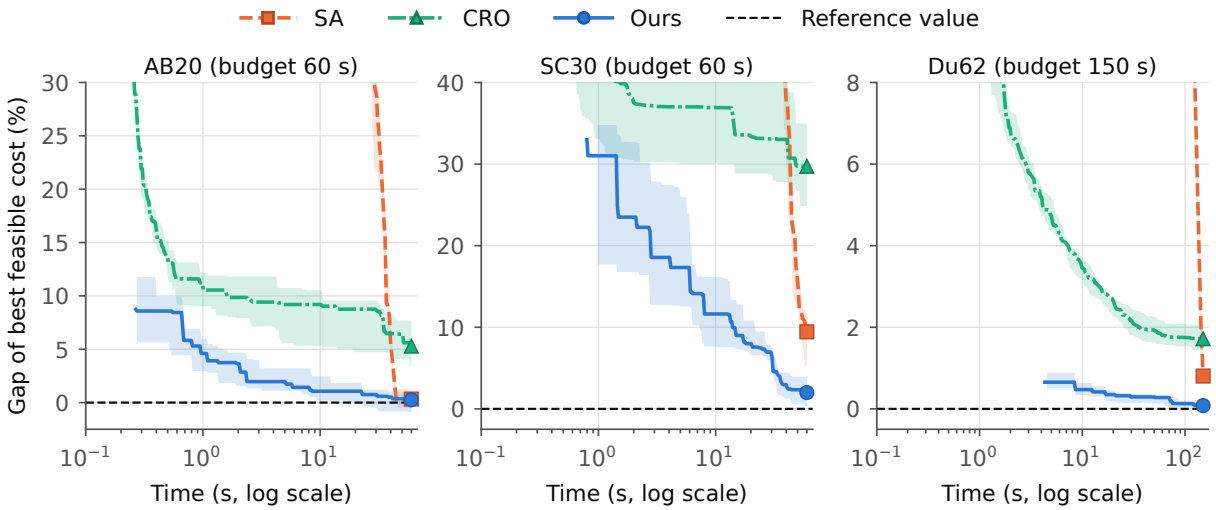


Figure 2: Anytime behaviour: gap of the best feasible cost found so far, median over 10 seeds (line) and interquartile range (band; drawn once all 10 runs have a feasible layout). The median is drawn once more than half of the runs have a feasible layout. Dashed line: reference value (AB20 5,021.17, SC30 3,343.47, Du62 3,635,307.30). The marker shows the median at the end of the budget. Swap and GA have no recorded traces in this data version and are not shown. The MINLP incumbent would be drawn as a separate step line, but it has no feasible layout on AB20 and Du62, and on SC30 its incumbent stays above the plotted range. Upper parts of the curves are cut off.

(adjusted $p = 1$) or Du62 (adjusted $p = 0.069$); the same holds against SA + polish (Table 3). Against Swap, GA, CRO and CRO + polish, the differences are significant on every instance where the costs differ. The construction heuristic is between 14.75 % (O7) and 195.51 % (SC30) above the references.

The polish control. One SOCP solve improves the SA layouts on O7 and SC30 in every run (median reduction 3.0 % on SC30) and changes almost nothing on the full floors (AB20: an improvement in 1 of 10 runs; Du62: median 0.0002 %). On SC30, SA + polish has mean gap 5.70 %, against 9.07 % for SA and 1.93 % for our method. So part of our advantage on SC30 comes from the SOCP geometry itself, and part from the search.

Anytime behaviour. Figure 2 shows the median best feasible cost over time. Our method has a feasible layout early and improves steadily. SA has a feasible layout in most runs only in the last part of the budget, and then improves fast. At the end of the budget its median gap is 0.33 % on AB20, 9.44 % on SC30 and 0.80 % on Du62, against 0.26 %, 2.00 % and 0.08 % for our method. Swap and GA are not shown because their runs were made with a code version that did not record the anytime trace.

Cost of Stage 2. The SOCP that evaluates one sequence pair took on average 0.6 ms on O7, 3.2 ms on AB20, 3.3 ms on SC30 and 64.4 ms on Du62. On the full floors many changed sequence pairs have no feasible geometry: 65 % of the SOCPs were infeasible on AB20, against 9 % on SC30, which has free area. On Du62 the budget rule applied in all 10 runs, so Stage 2 was a single polish there. As a consequence, the full method and the two Stage-2 ablations (Section 2.6) gave identical results on Du62 in every seed.

2.5 Best layouts and published values

Figure 3 shows the best layout of our method for each instance and Table 4 compares these layouts with the best published values. The best layouts come from longer, separately configured runs (up to 900 s, with other settings of the time split and the elite pool); they are not protocol results. The column “Best protocol run” gives the lowest cost over all protocol runs of all methods. All layouts were checked again by the evaluator.

Table 4: Best published values and our best layouts. Area rule: *exact* means every department meets its area exactly; *outer approx.* means the area constraint $w_i h_i \geq A_i$ is replaced by tangent cuts, so areas can be slightly under-met and the value can lie below any exact-area layout. Comparable (Cond.: conditional): whether the value can be compared with a strictly feasible layout on our instance file. Best protocol run: lowest cost over all protocol runs of all methods (method in brackets). Ours (record): best layout of our method from longer, separately configured runs (up to 900 s; not protocol results), re-checked by the evaluator. Δ: Ours (record) relative to the published value. G&R: Gonçalves and Resende; J&C: Jeong and Chae; K&C: Kang and Chae.

Inst.	Published	Source	Area rule	Comparable	Best protocol run	Ours (record)	Δ (%)
O7	131.6531381	MINLPLib o7 [2]	exact	Yes: proven optimum	131.6531 (MINLP)	131.6532	0.000
	131.56	G&R, T. 10 [14]	outer approx.	No: below the optimum			0.071
	131.6773	Hou 2019[a]	unknown	Above the optimum			−0.018
	131.69	J&C, T. 6 [1]	exact	Yes; best slicing layout			−0.028
vC10a	18,522.79	J&C, T. 6 [1]	exact	Yes, $w_{9,10} = 59.2$[b]	18,522.79 (tie of 8 methods)	18,522.79	0.00
AB20	4,927.27	J&C, T. 6 [1]	exact	No: aspect 4.008 $> \alpha$	4,861.35 (Ours, elite pool 1)	4,861.35	−1.34
	5,021.17	G&R, T. 12 [14]	outer approx.	Cond.: flow version not recorded			−3.18
SC30	3,343.47	J&C, T. 6 [1]	exact	Yes, in our favour[c]	3,301.27 (Ours)	3,279.66	−1.91
	3,367.87	G&R, T. 12 [14]	outer approx.	Cond.: flow version not recorded			−2.62
	3,412.87	Hou 2019[a]	unknown	Cond.: rules not recorded			−3.90
Du62	3,635,307.30	K&C via J&C [9]	unknown	Cond.: floor unconfirmed[d]	3,623,751.81 (SA + polish)	3,615,501.31	−0.54
	3,672,799.99	J&C, T. 6 [1]	exact	Yes (117.124 square)			−1.56

[a] Value as recorded in our literature notes; floor, aspect rule and area handling of this source were not checked, and the full reference is not verified.

[b] Our instance file uses $w_{9,10} = 59.2$, the value under which the published layout gives 18,522.79.

[c] The published layouts are slicing layouts that fill the free area with 17 dummy blocks; our model allows free space anywhere and non-slicing layouts, so it is less restrictive.

[d] Secondary citation (Jeong and Chae; Kang13 data file); the floor is probably the 117.124 square but this is not confirmed from the original paper.

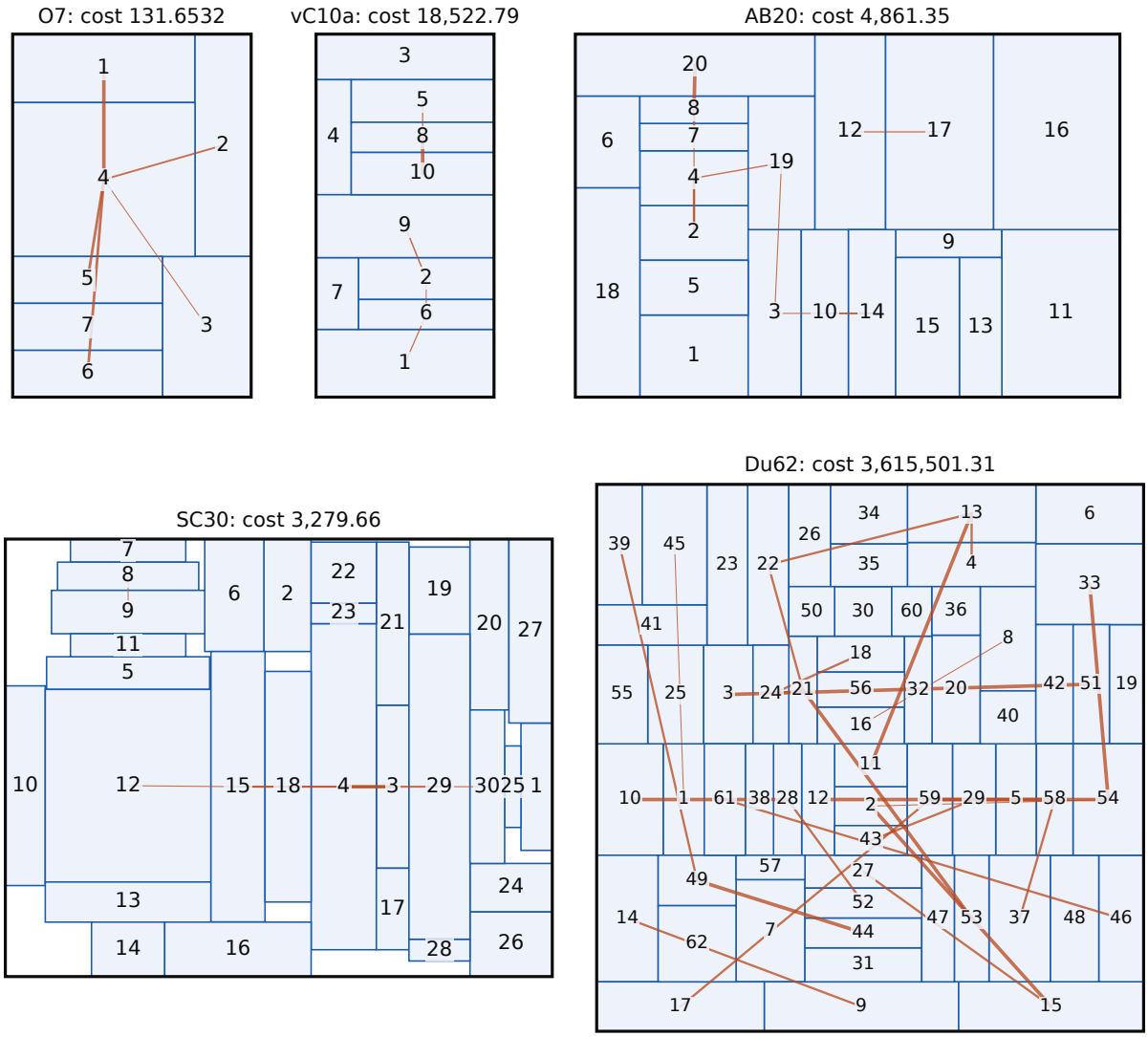


Figure 3: Best layouts of our method (record runs, not protocol runs), drawn to scale within each panel (thick outline: floor). Numbers are department indices; lines join the centroids of the department pairs with the largest flows (top 15 % of the pairs with non-zero flow, at least 5 and at most 40; 25 for Du62), with line width growing with the flow. White areas in SC30 are free floor. Costs: O7 131.6531781, vC10a 18,522.79, AB20 4,861.35, SC30 3,279.66, Du62 3,615,501.31.

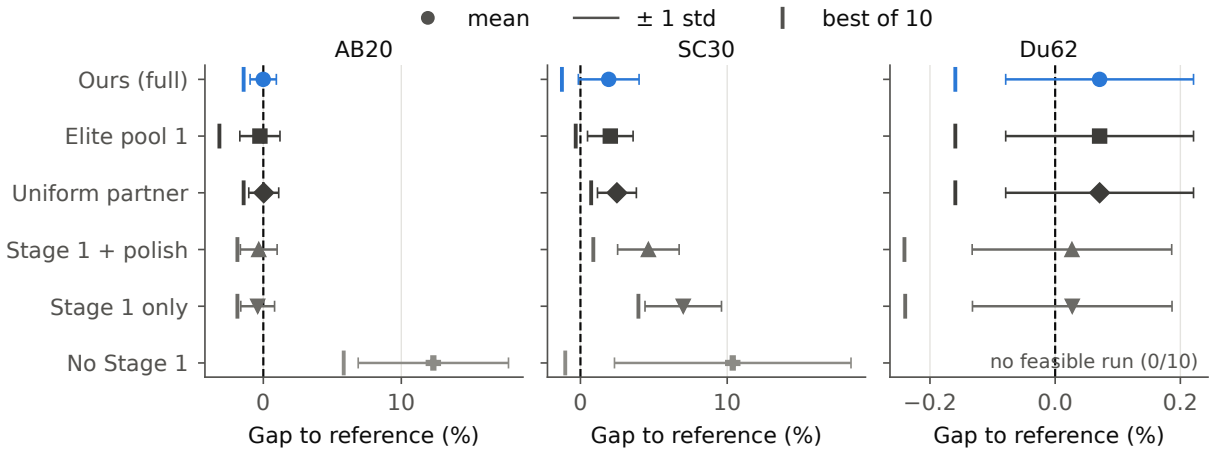


Figure 4: Ablation on AB20, SC30 and Du62: mean gap (± one standard deviation, 10 seeds) and best gap of the full method and five variants. References as in Figure 1. Note the different horizontal scales.

Published values were obtained under different rules, so we compare only where the rules match, and we report the others as they are.

- **O7: no record is possible.** The instance is solved to proven optimality (131.6531381). The value 131.69 of Jeong and Chae [1] is the best slicing layout. The value 131.56 of Gonçalves and Resende [14] lies below the proven optimum, so their layout cannot meet every area exactly; it uses a tangent outer approximation of the area constraint. Our layout costs 131.6531781, which is the optimum up to our area safety margin (relative difference $3.0 \cdot 10^{-7}$).
- **vC10a.** Our best layout is the published layout of [1] and has the same cost, 18,522.79. In the protocol runs it is reached by 8 methods (a tie, not a record).
- **AB20.** Our best layout (4,861.35) is 3.18 % below the value of Gonçalves and Resende [14], which uses the area outer approximation and a flow-matrix version that is not recorded. The lower published value 4,927.27 [1] belongs to a layout in which two departments have aspect ratio 4.008, above $\alpha = 4$, so it is not feasible under the strict limit. The plain SA baseline also goes below that value (best 4,926.92). No published AB20 value that we know is both exact-area and strictly feasible.
- **SC30.** Our best layout (3,279.66) is 1.91 % below the best value of Jeong and Chae [1]. The comparison is in our favour: their layouts are slicing layouts that fill the free area with dummy blocks, while our model allows non-slicing layouts and free space anywhere. Every layout of theirs is feasible for us, but not the other way round.
- **Du62.** Our best layout (3,615,501.31, one long Stage-1 anneal and one SOCP polish) is 1.56 % below the value of Jeong and Chae [1] on the same square floor, and 0.54 % below the value cited for Kang and Chae [9]. That value is a secondary citation, and its floor is not confirmed. The plain SA baseline also goes below both values in a protocol run (best 3,623,772.52, 1.33 % and 0.32 % below them), so the improvement over the literature on Du62 comes from slicing-tree annealing, not from Stage 2.

2.6 Ablation

Table 5 and Figure 4 compare the full method with five variants under the same protocol, and Figure 5 shows how much Stage 2 lowers the cost within each run.

Stage 1 is needed. Without Stage 1, Stage 2 starts from a random feasible sequence pair. The mean gap rises to 12.32 % on AB20 and 10.38 % on SC30, and on Du62 no random feasible sequence pair was found in any run (0/10).

Table 5: Ablation of the two-stage method (10 seeds, same budgets as Table 2). Stage 1 only: slicing-tree annealing for the whole budget; Stage 1 + polish: one SOCP solve on the final Stage-1 layout; No Stage 1: Stage 2 started from a random feasible sequence pair; Elite pool 1: Stage 2 from the best Stage-1 layout only; Uniform partner: Stage-2 move partner chosen uniformly ($p_{\text{near}} = 0$). The last three variants were run on AB20, SC30 and Du62 only. Bold: lowest mean per instance among variants with all runs feasible.

Variant	Feas.	Best	Mean ± std	Gap mean (%)
O7 (budget 20 s)				
Ours (full)	10/10	131.65	**131.65** ± 0.00	0.00
Stage 1 + polish	10/10	131.65	131.66 ± 0.00	0.00
Stage 1 only	10/10	131.69	131.69 ± 0.00	0.03
vC10a (budget 20 s)				
Ours (full)	10/10	18,522.79	**18,522.79** ± 0.00	0.00
Stage 1 + polish	10/10	18,522.79	**18,522.79** ± 0.00	0.00
Stage 1 only	10/10	18,522.79	**18,522.79** ± 0.00	0.00
AB20 (budget 60 s)				
Ours (full)	10/10	4,949.54	5,020.81 ± 47.79	−0.01
Elite pool 1	10/10	4,861.35	5,008.76 ± 73.04	−0.25
Uniform partner	10/10	4,949.54	5,022.71 ± 54.48	0.03
Stage 1 + polish	10/10	4,926.92	5,004.52 ± 66.91	−0.33
Stage 1 only	10/10	4,926.92	**5,000.64** ± 61.49	−0.41
No Stage 1	10/10	5,313.70	5,639.86 ± 273.43	12.32
SC30 (budget 60 s)				
Ours (full)	10/10	3,301.27	**3,407.85** ± 69.18	1.93
Elite pool 1	10/10	3,332.69	3,411.51 ± 51.91	2.04
Uniform partner	10/10	3,367.91	3,426.57 ± 44.31	2.49
Stage 1 + polish	10/10	3,372.80	3,498.23 ± 70.07	4.63
Stage 1 only	10/10	3,475.55	3,577.69 ± 87.12	7.01
No Stage 1	10/10	3,308.88	3,690.44 ± 269.44	10.38
Du62 (budget 150 s)				
Ours (full)	10/10	3,629,508.93	3,637,888.92 ± 5,454.26	0.07
Elite pool 1	10/10	3,629,508.93	3,637,888.92 ± 5,454.26	0.07
Uniform partner	10/10	3,629,508.93	3,637,888.92 ± 5,454.26	0.07
Stage 1 + polish	10/10	3,626,553.92	**3,636,281.42** ± 5,794.02	0.03
Stage 1 only	10/10	3,626,598.70	3,636,299.54 ± 5,794.90	0.03
No Stage 1	0/10	–	–	–

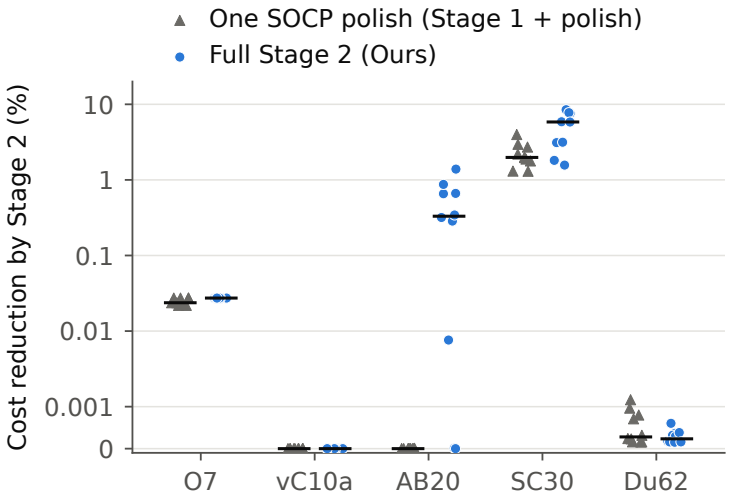


Figure 5: Cost reduction by Stage 2 within each run, $100\,(c_1 - c)/c_1$, where c_1 is the best Stage-1 cost of the run and c the final cost. Triangles: Stage 1 + polish (one SOCP solve on the final Stage-1 layout); circles: full method (Stage 1 for half of the budget, then Stage 2). Bars: medians. Symmetric log scale, linear below 10^{-3} %. On Du62 the full method also reduces to a polish (budget rule).

Stage 2 matters on SC30 and O7 only. On SC30 a single polish lowers the cost by a median of 1.98 % and the full Stage 2 by 5.84 % (Figure 5); the mean gap falls from 7.01 % (Stage 1 only) to 4.63 % (polish) and 1.93 % (full). The full method is significantly better than both variants on SC30 (adjusted $p = 1.8 \cdot 10^{-3}$ and 0.023, Table 3). On O7 a single polish already turns the best slicing layout into the optimal non-slicing layout. On the full floors the effect is small or absent. On vC10a and AB20 the polish lowers the cost by less than 10^{-5} % in every run, because a fixed sequence pair on a full floor fixes the geometry. On AB20 the Stage-2 search lowers the cost by a median of 0.331 % relative to the Stage-1 result at half time, but the full method (mean 5,020.81) is not better than Stage 1 alone run for the whole budget (mean 5,000.64; adjusted $p = 1$). On Du62 Stage 2 is a single polish.

Elite pool and neighbour moves. With an elite pool of size one, the mean is 5,008.76 on AB20 and 3,411.51 on SC30, against 5,020.81 and 3,407.85 for the full method; with uniform partner choice it is 5,022.71 and 3,426.57. These differences are within one standard deviation, so ten seeds do not show an effect of either component. The best AB20 layout of all protocol runs (4,861.35) was found by the variant with elite pool one.

2.7 Threats to validity

- **Tuning on test instances.** The Stage-1 parameters of our method were tuned on AB20 and SC30, and the Stage-2 temperatures on SC30; these are benchmark instances. The baselines were not tuned. The best layouts in Table 4 come from longer runs with other settings and are not protocol results.
- **Wall-clock budgets.** Budgets are in seconds on one machine. Results depend on the machine and on the load from the other two parallel runs, and runs are not bit-reproducible. The runs of our method overran the budget by up to 2.9 % on Du62 (Section 2.3); the baselines did not. The exact models ran on a different machine with two cores.
- **Baselines.** Swap, GA and SA are our own implementations on the STS encoding, with parameters chosen by us; GA has no local search. CRO is our reproduction, run under the common budget instead of the paper's stopping rule and with a static penalty on SC30; it does not reproduce the published SC30 results even with the full stopping rule. Stronger baselines may exist.
- **Exact model.** The MINLP got one 300 s run per instance; with more time or a stronger formulation it may do better, as the 3600 s O7 run shows.

- **Reference values.** Several published values use other area rules, other data versions or an unconfirmed floor (Table 4); gaps to them are indicative only. The reference values were checked against our literature notes, not against the original papers in this study.
- **Sample size.** Ten seeds per method and instance give limited power; the AB20 and Du62 differences to SA and most ablation differences are not significant.
- **Tolerance.** The evaluator accepts violations up to 10^{-6}. Solver-based layouts (SOCP, MINLP) meet the limits to about 10^{-7} (the best AB20 layout has a largest residual of order 10^{-10}, since a full floor leaves no room for a safety margin); slicing layouts are exact to about 10^{-9}.

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